Activation Functions

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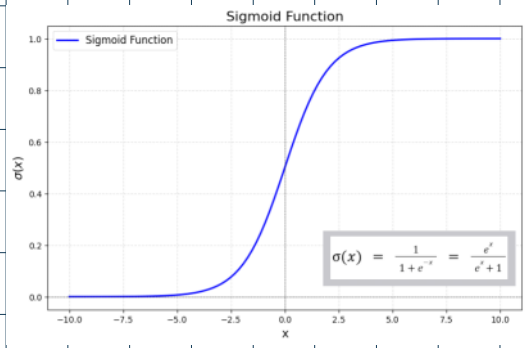
The core idea behind activation functions is introducing non linearity to our machine learning algorithm so that it can mimic any curve. without activation functions, our model can only learn linear relationships, which is not very useful.

Sigmoid:

$$S(x) = \frac{1}{1+e^{-x}}$$



Sigmoid derivative:

$$\frac{(1)'(1+e^{-x}) - 1\cdot(1+e^{-x})'}{(1+e^{-x})^2} \Rightarrow \frac{e^{-x}}{(1+e^{-x})^2}$$

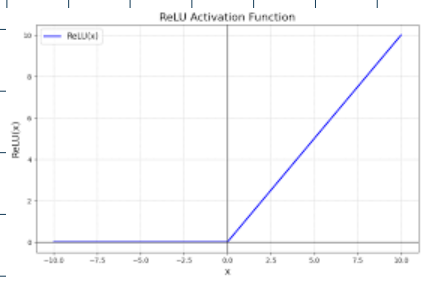
Notice $1 - S(x) = \frac{e^{-x}+1}{e^{-x}+1} - \frac{1}{e^{-x}+1} = \frac{e^{-x}}{e^{-x}+1}$

then, $[1-S(x)]\cdot S(x) = \frac{e^{-x}}{e^{-x}+1}\cdot\frac{1}{e^{-x}+1} = \frac{e^{-x}}{(e^{-x}+1)^2} = S'(x)$

$$S'(x) = [1-S(x)]\cdot S(x)$$

ReLU:

$\max(0,x)$



ReLU derivative:

0 if $x<0$

1 if $x>0$

That's literally it. You may wonder, how does a seemingly linear function even introduce non-linearity? which is a totally valid question. Here is a link to a great video explaining it below:

https://www.youtube.com/watch?v=68BZ5f7P94E

Swish:

$$\text{swish}(x) = x \cdot \text{sigmoid}(x)$$

Swish derivative:

$$x' \cdot \text{sigmoid}(x) + x \cdot \text{sigmoid}'(x) = \text{sigmoid}(x) + x \cdot \text{sigmoid}(x) \cdot (1 - \text{sigmoid}(x))$$

$$\text{swish}'(x) = \text{sigmoid}(x) + x \cdot \text{sigmoid}(x) \cdot (1 - \text{sigmoid}(x))$$

Swish is just a smoother alternative to ReLU